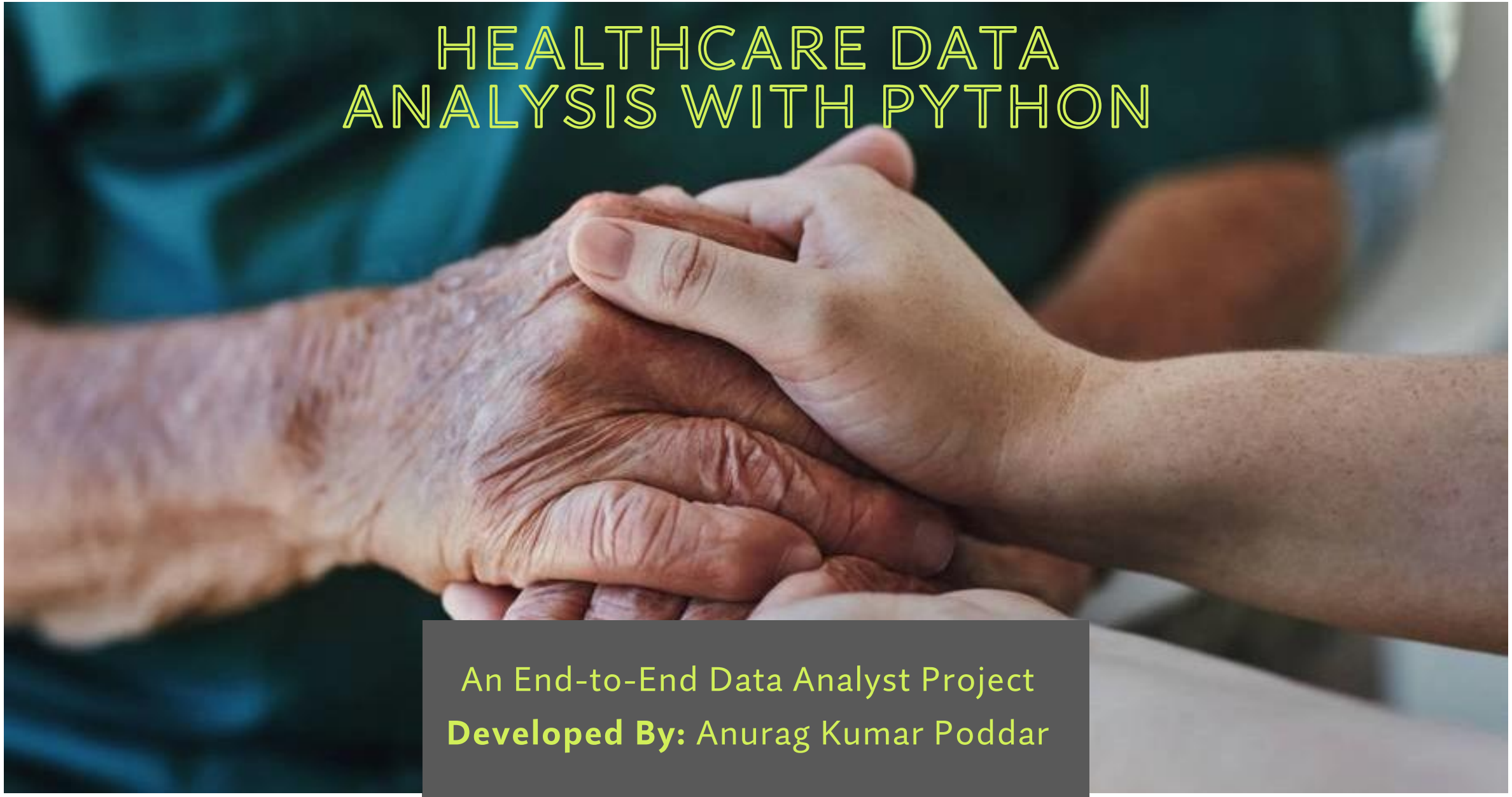
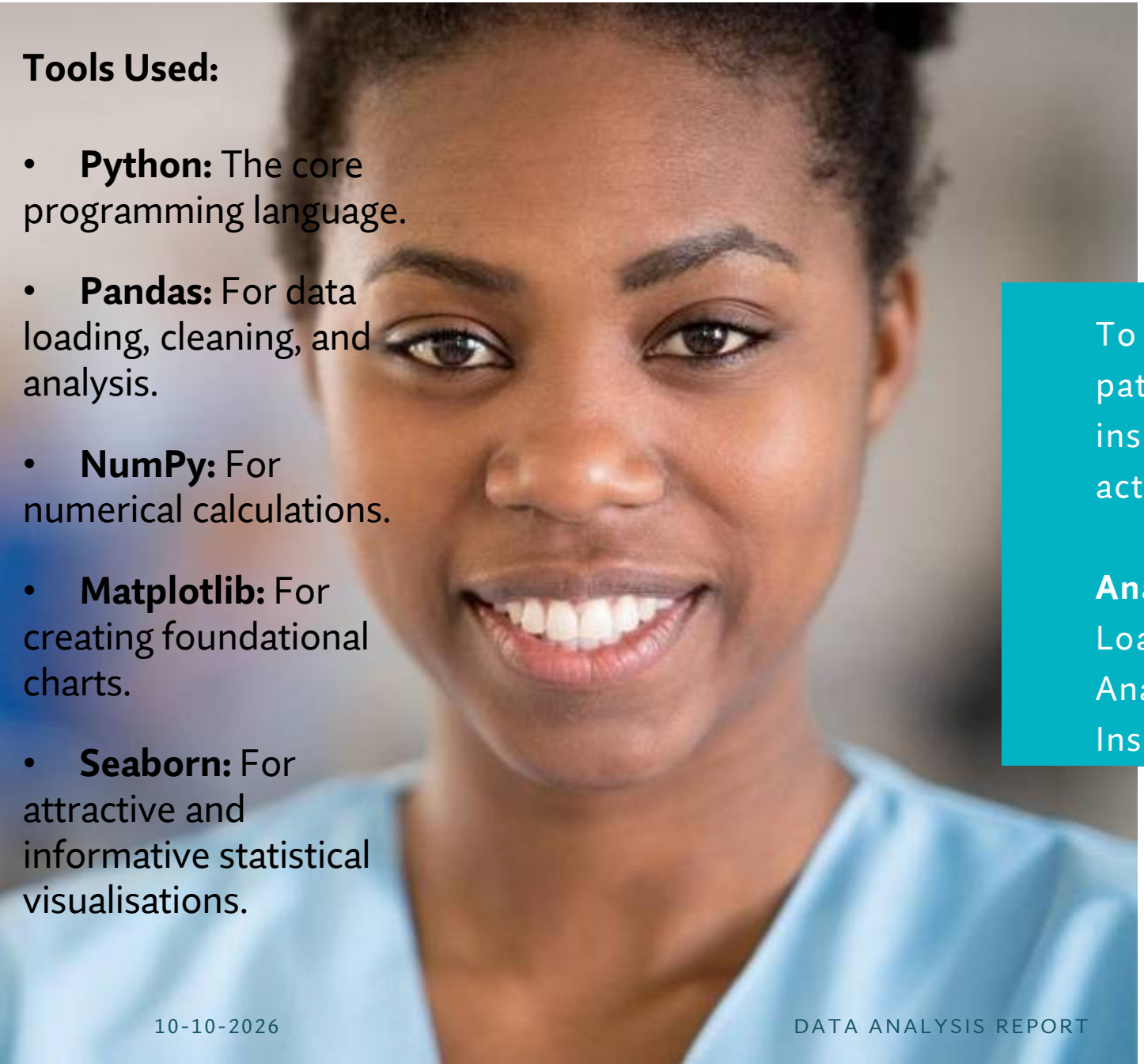


HEALTHCARE DATA ANALYSIS WITH PYTHON

An End-to-End Data Analyst Project

Developed By: Anurag Kumar Poddar



Tools Used:

- **Python:** The core programming language.
- **Pandas:** For data loading, cleaning, and analysis.
- **NumPy:** For numerical calculations.
- **Matplotlib:** For creating foundational charts.
- **Seaborn:** For attractive and informative statistical visualisations.

PROJECT GOAL

To analyse a healthcare dataset containing patient, admission, medical condition, billing, insurance, and test-result information to uncover actionable business insights.

Analysis Workflow:
Load Data → Understand Data → Clean Data → Analyse Data → Visualise Results → Find Business Insights

DATA LOADING & INITIAL INSPECTION

KEY DATA COLUMNS IDENTIFIED:

Patient Info: Name, Age, Gender, Blood Type
Admission Details: Date of Admission, Doctor, Hospital, Admission Type
Medical Info: Medical Condition, Medication, Test Results
Financials: Billing Amount, Insurance Provider

Medical equipment sales were up by $25 million in 2020

DATA SOURCE:
HEALTHCARE_DATASET.CSV

INITIAL SIZE: 55,500 ROWS AND 15 COLUMNS.

INITIAL OBSERVATIONS:

The dataset contained no missing values.
Data types were a mix of numerical and categorical (object) data.
Initial statistical summary revealed some key metrics (e.g., Average Age: 51.5, Average Billing: 25,539).



DATA CLEANING & FEATURE ENGINEERING

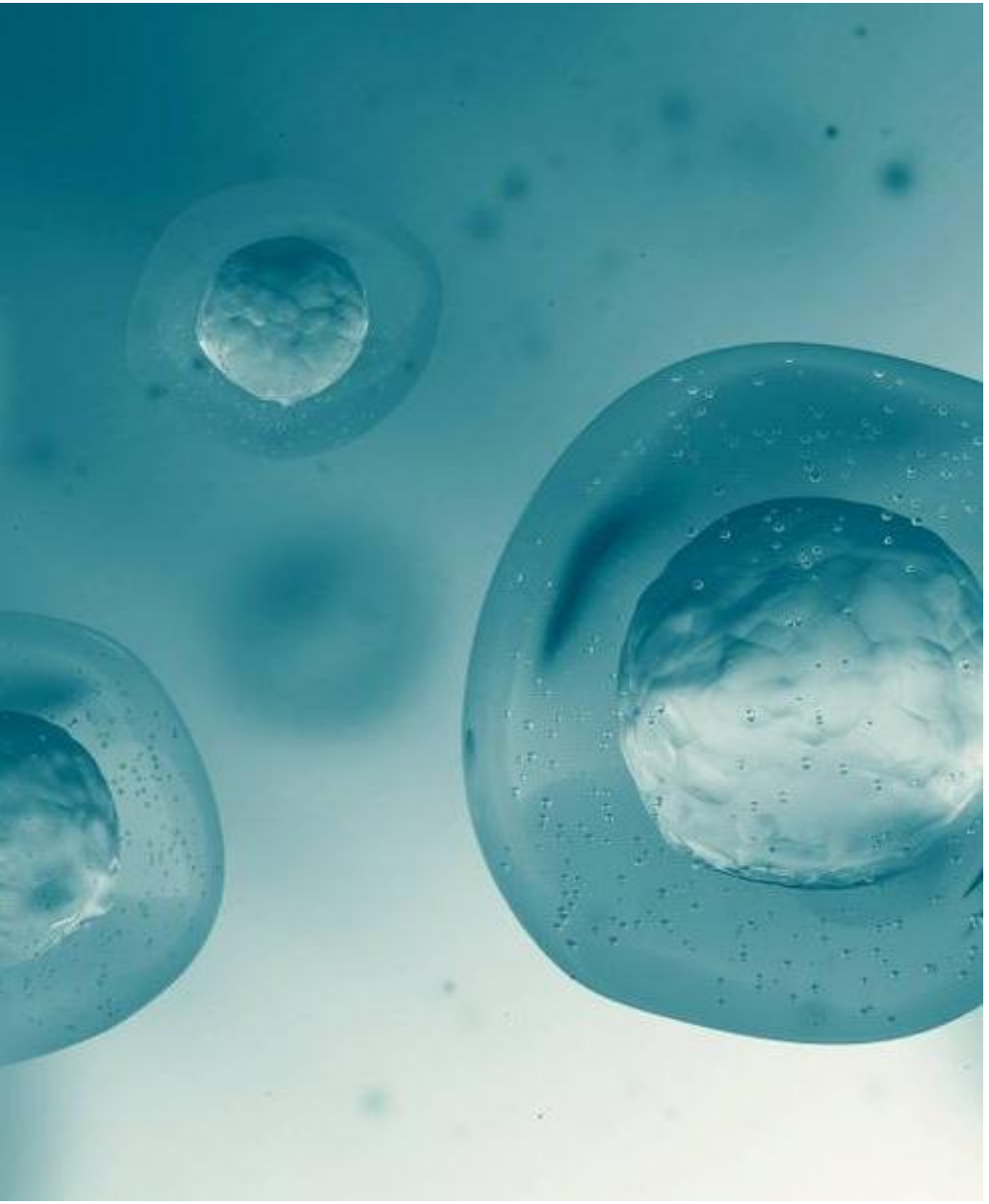
1. DATA CLEANING:

Removed Duplicates: 534 duplicate rows were identified and removed.
Converted Data Types: 'Date of Admission' and 'Discharge Date' were converted from text to proper datetime format.
Handled Invalid Billing: 106 records with negative billing amounts were identified and replaced with the median of valid billing amounts (25,593.87).

2. FEATURE ENGINEERING (CREATED NEW COLUMNS):

Length of Stay: Calculated as Discharge Date - Admission Date.
Age Group: Categorised patients into groups (0-18, 19-35, 36-50, 51-65, 66+).
Admission Month: Extracted month and year from the admission date for time-series analysis.

FINAL CLEAN DATASET: 54,966 ROWS AND 17 COLUMNS.



EXPLORATORY DATA ANALYSIS (EDA) - KEY METRICS

Overall Patient Statistics:

- **Total Patients (after cleaning):** 54,966
- **Average Age:** 51.54 years
- **Average Billing Amount:** 25,594.63
- **Average Length of Stay:** 15.50 days

Most Common Categories:

- **Medical Condition:** Arthritis (9,218 patients)
- **Admission Type:** Elective (18,473 patients)
- **Test Result:** Abnormal (18,437 patients)
- **Insurance Provider:** Cigna (11,139 patients)

These metrics provide a high-level summary of the patient population and hospital operations.

VISUAL ANALYSIS - PATIENT DEMOGRAPHICS

Patient Distribution by Gender

The patient population is almost perfectly balanced, with Male (27,496) and Female (27,470) counts being nearly identical.

Distribution of Patient Age

The age of patients follows a relatively normal distribution, centered around the average of 52 years.

The largest patient group falls into the 51-65 and 66+ age brackets.

VISUAL ANALYSIS - CLINICAL & OPERATIONAL INSIGHTS

Number of Patients by Medical Condition

- Patient counts are very evenly distributed across all six medical conditions.
- Arthritis and Diabetes are slightly more common, but no single condition dominates.

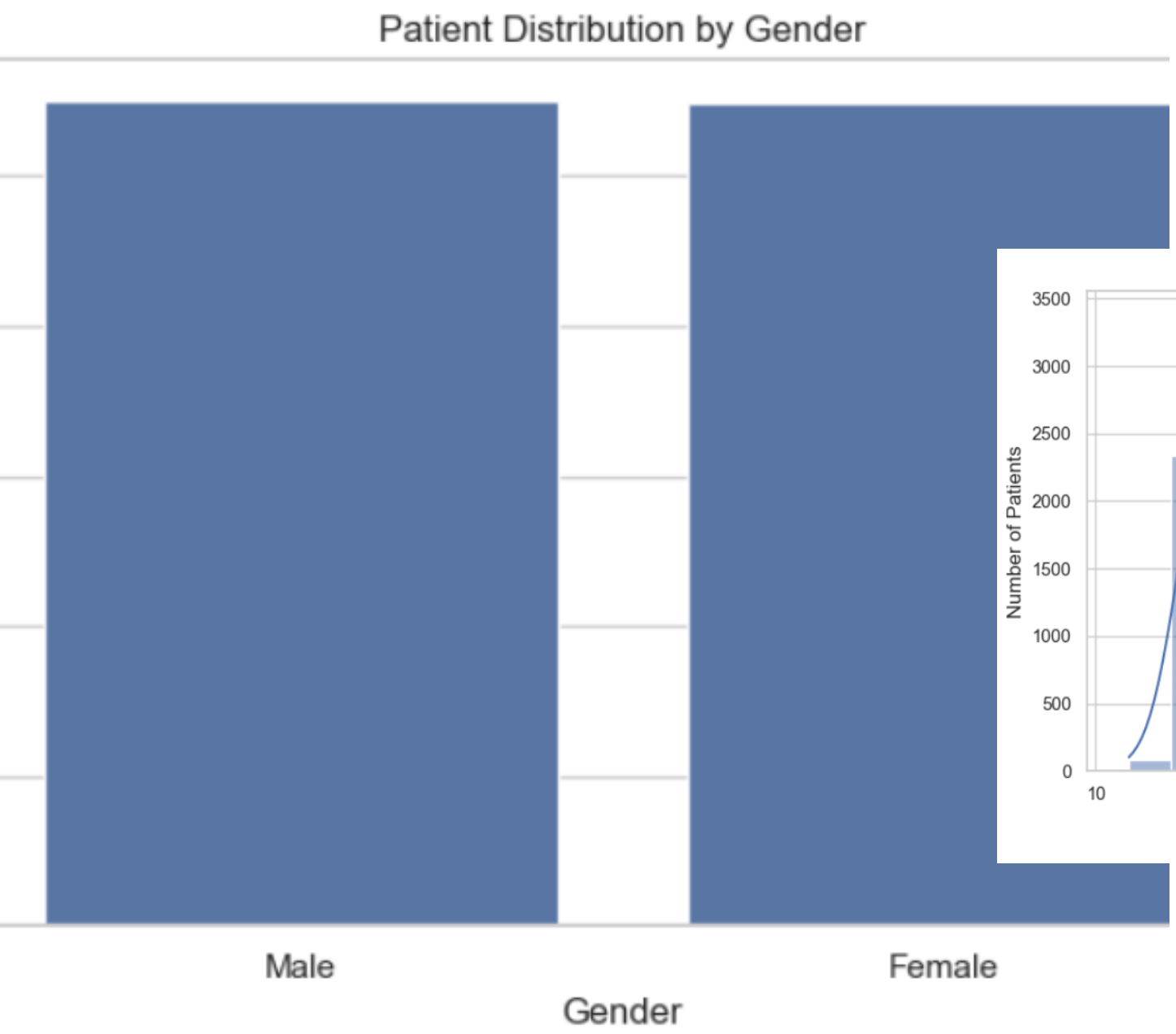
Test Results by Gender

- Test result outcomes (Normal, Abnormal, Inconclusive) are distributed evenly between male and female patients.

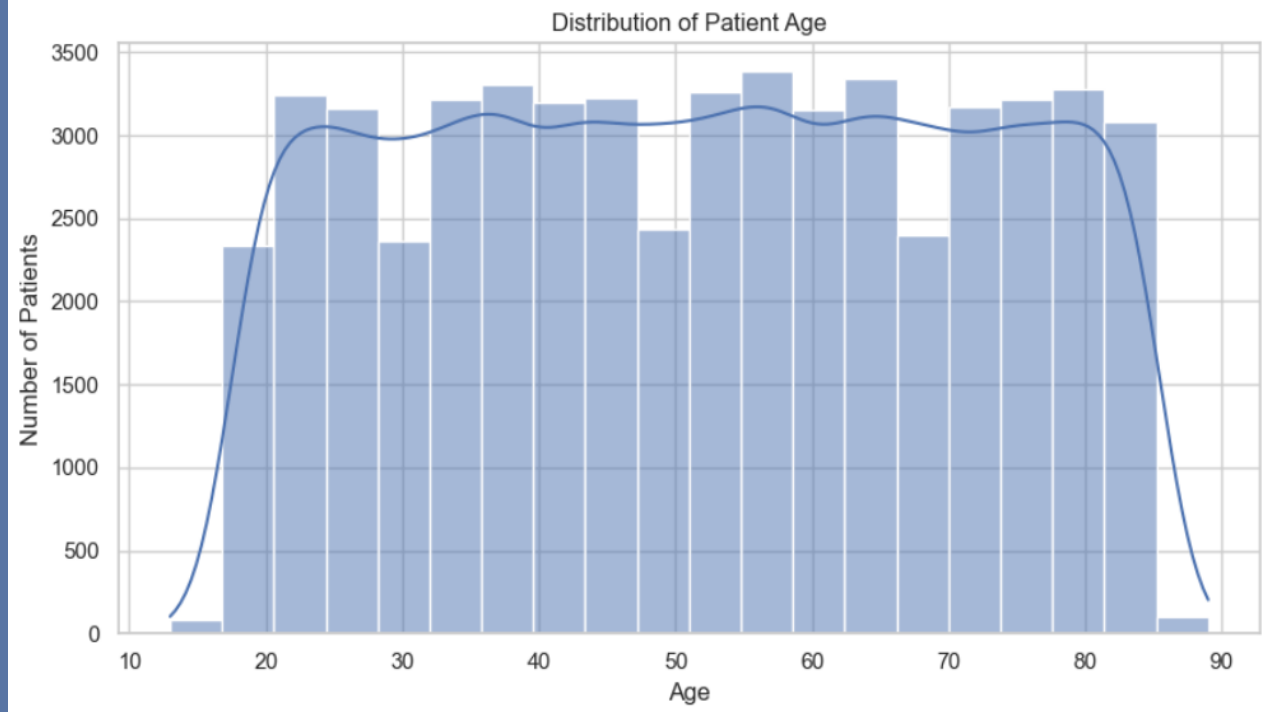
Patients by Admission Type

- Elective, Urgent, and Emergency admissions are almost equally common.
- This suggests a consistent and varied case mix for the hospital.

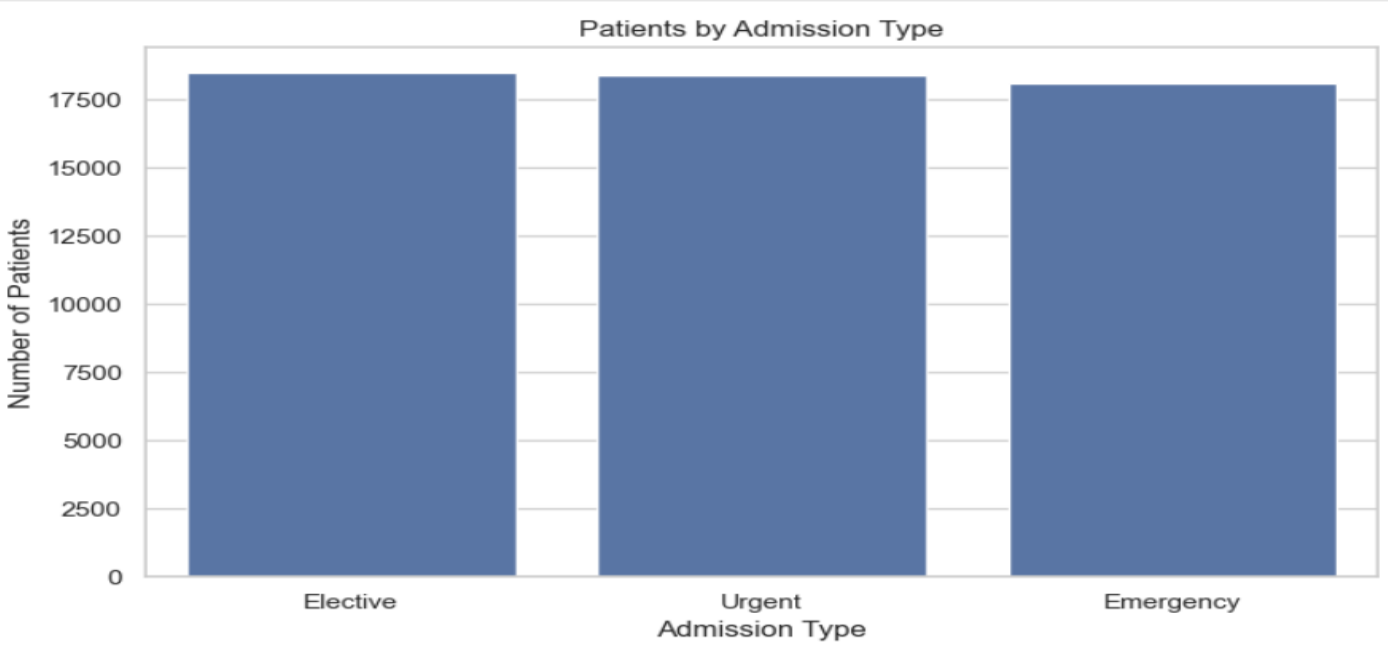
PATIENT DISTRIBUTION BY GENDER

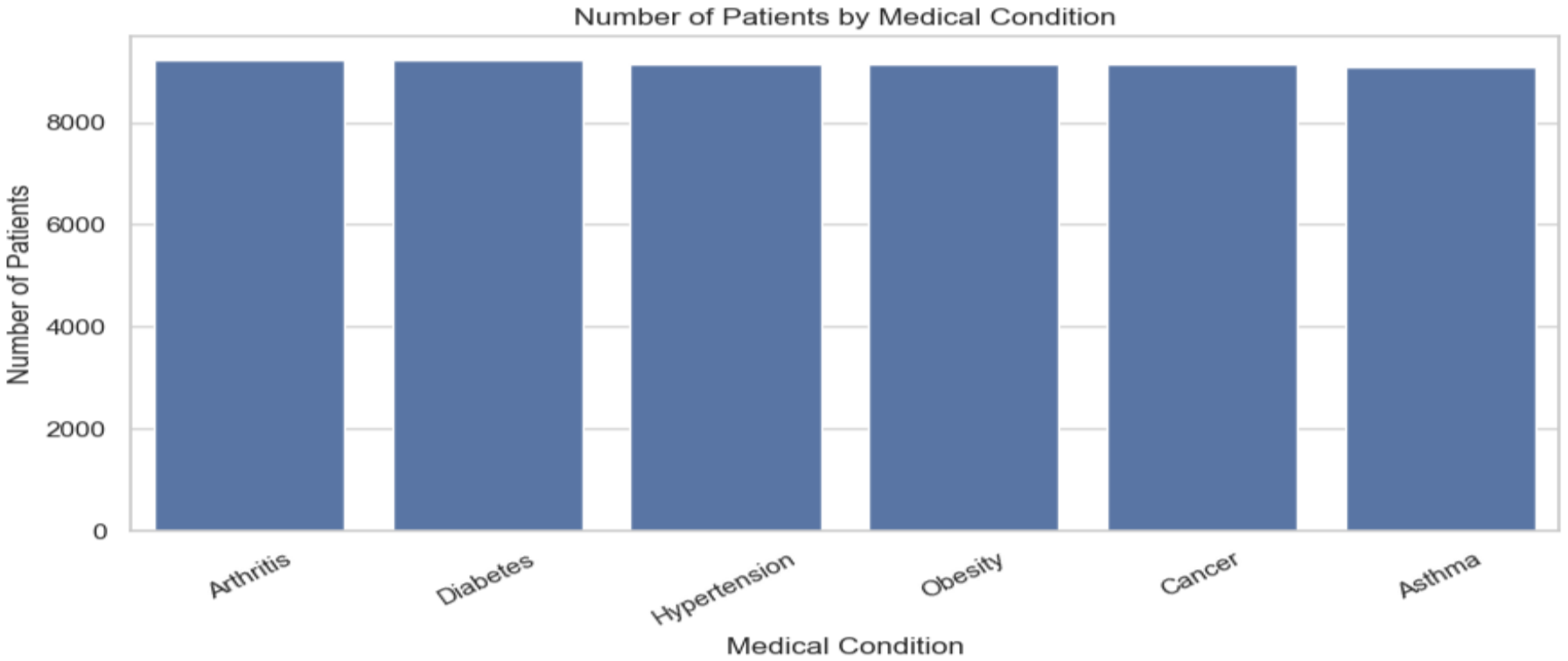


DISTRIBUTION OF PATIENT AGE

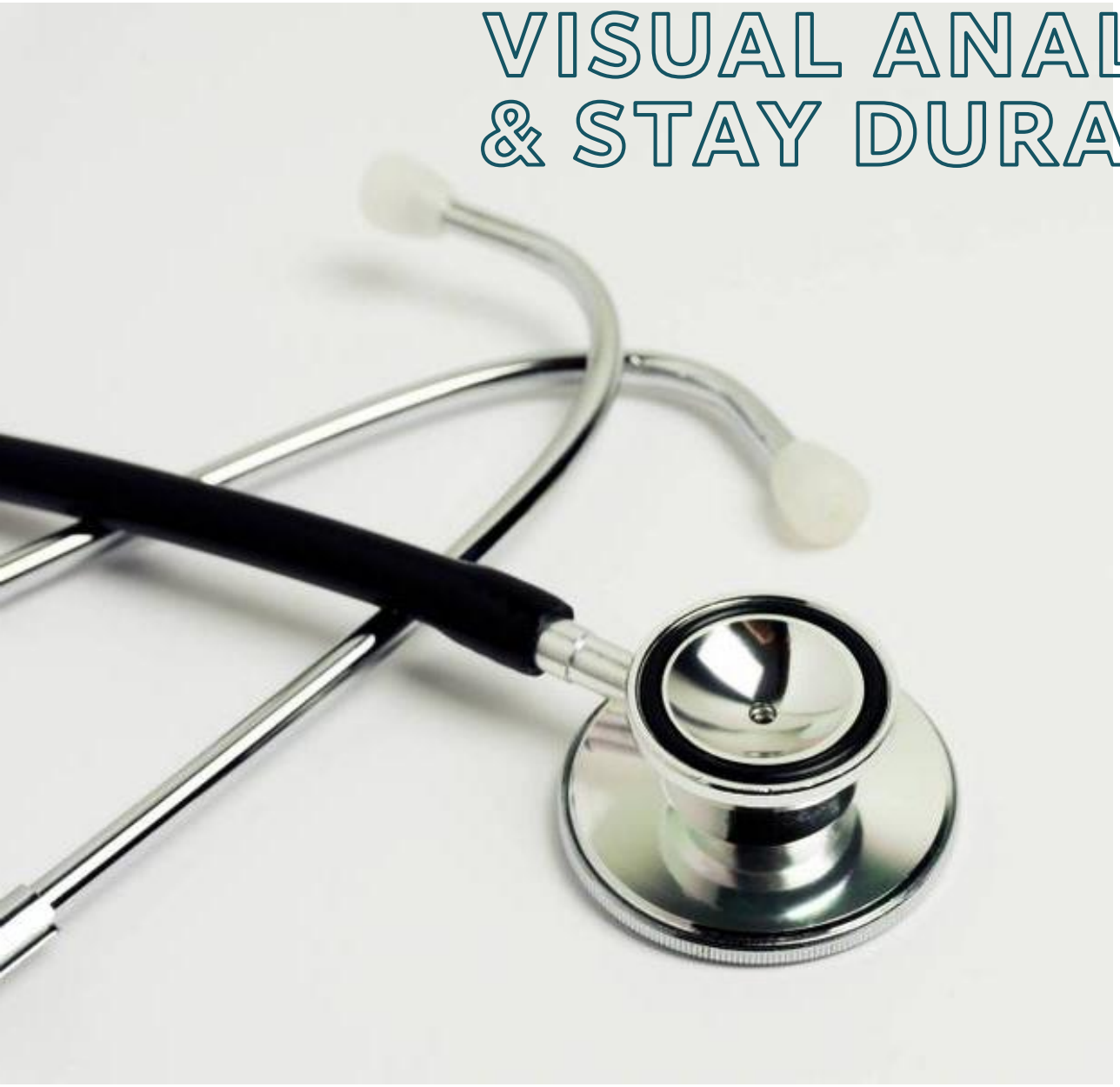


CLINICAL & OPERATIONAL INSIGHTS





VISUAL ANALYSIS - FINANCIAL & STAY DURATION



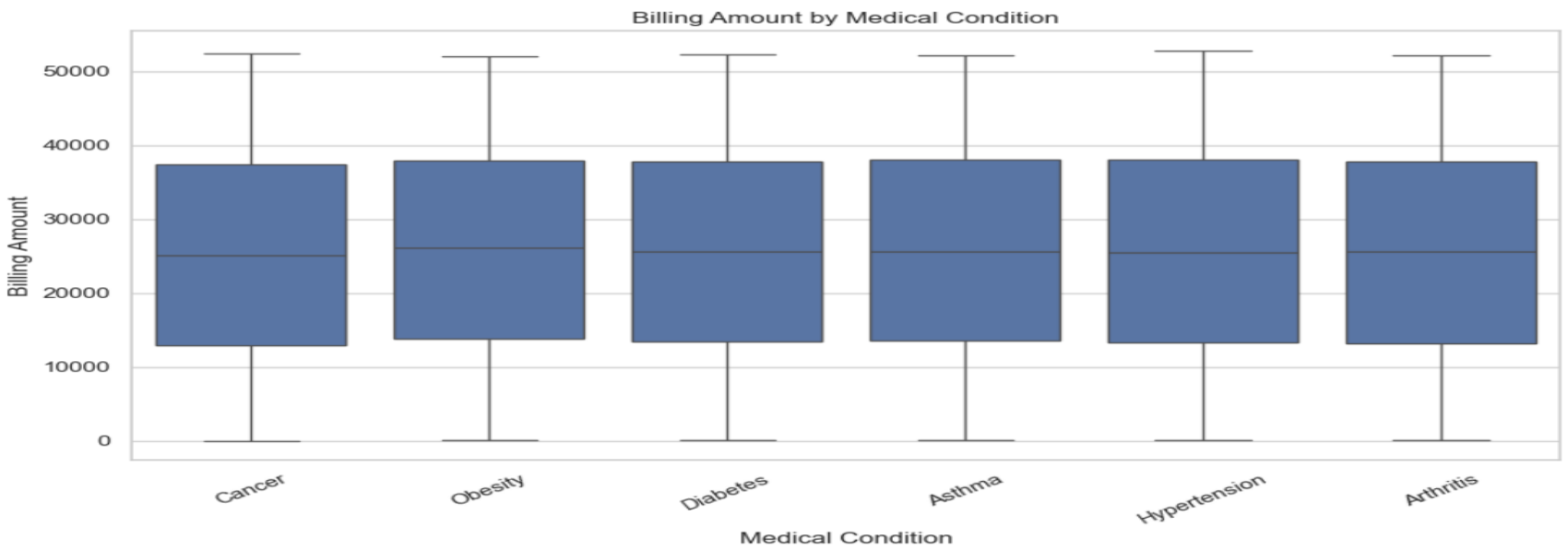
BILLING AMOUNT BY MEDICAL CONDITION

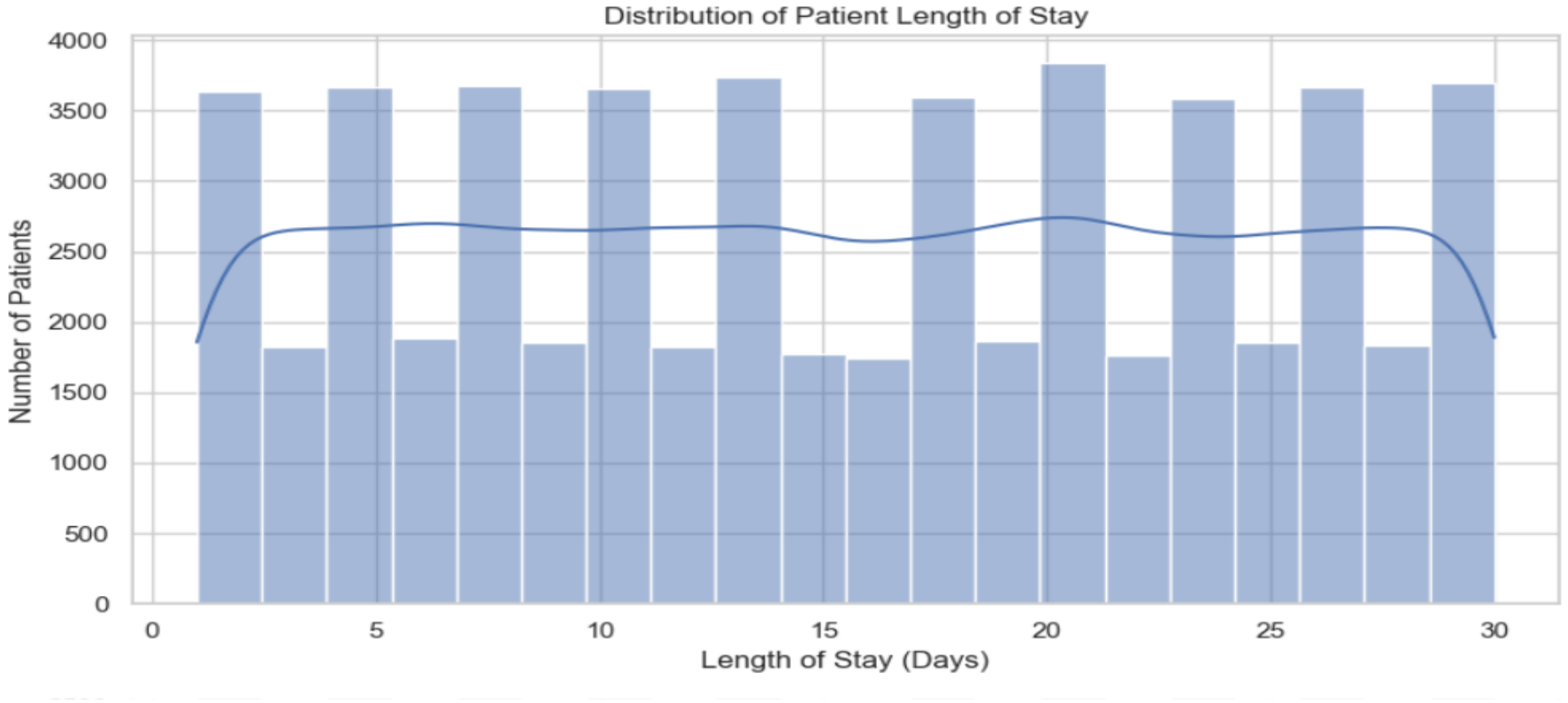
The average billing amount is very consistent across all medical conditions, hovering around the overall mean of ~25,600.
Obesity has the highest average billing, while Cancer has the lowest, but the difference is minimal.

DISTRIBUTION OF PATIENT LENGTH OF STAY

The length of stay is fairly evenly distributed, with a slight skew towards longer stays.
The average stay is approximately 15.5 days across all conditions.

FINANCIAL & STAY DURATION





KEY FINDINGS & BUSINESS INSIGHTS



KEY FINDINGS:

Most Common Condition: Arthritis (9,218 patients).

Highest Average Billing Condition: Obesity (₹25,858.66).

Most Common Admission Type: Elective.

Most Common Test Result: Abnormal.

Overall Averages: Age (51.54 years), Billing (₹25,594.63), Stay (15.50 days).

BUSINESS INSIGHTS:

Balanced Case Mix: The hospital handles a diverse range of conditions and admission types, indicating a stable operational load.

Consistent Billing: Billing is uniform across conditions, suggesting standardised treatment protocols or pricing structures.

Data-Driven Decisions: The cleaned data can be used to optimise resource allocation (e.g., bed management for an average 15.5-day stay) and investigate specific cost drivers for conditions like Obesity.

OUR COMPETITION

CONTOSO

Our product is priced below that of other medical supply companies on the market

Quick and easy to use, compared to the complex ordering system of the competitors

Affordability is the main draw for our hospitals to our product

COMPETITORS

Company A
Product is more expensive

Companies B & C
Product is expensive and inconvenient to use

Companies D & E
Product is affordable, but inconvenient to use



CONCLUSION

This project successfully demonstrates an end-to-end data analysis workflow, from raw data cleaning to generating actionable business insights. The analysis provides a clear, high-level overview of patient demographics, operational patterns, and financial metrics within the healthcare dataset.

SKILLS DEMONSTRATED

- **Data Wrangling:** Pandas, NumPy
- **Data Cleaning:** Handling Duplicates, Invalid Values, and Type Conversion
- **Exploratory Data Analysis (EDA):** Statistical Summaries, Grouping, and Aggregation
- **Data Visualisation:** Matplotlib, Seaborn
- **Business Acumen:** Translating data into insights and recommendations

THANK YOU



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